4.8 Slicing a Spherical Loaf

WORKSHEET

The question

STEP I 2001, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The sphere of radius a is obtained by rotating which curve about the x-axis? What is $\frac{dy}{dx}$ for that curve?

STAGE 1 The crust formula simplifies

Covers: the crust is proportional to the number of slices

Ask yourself Which curve, rotated about the x-axis, gives the sphere of radius a? What is $\frac{dy}{dx}$ for it?

Pause and try Work out $y\sqrt{1+(y')^2}$ for that curve. Something remarkable happens.

STAGE 2 Crust is proportional to slices

Covers: the crust is proportional to the number of slices

Ask yourself So what is the crust area of a slice of thickness h? Does it depend on where the slice is?

STAGE 3 The crusty condition

Covers: the condition on the radius

Ask yourself Follow the hint: eat slices from the end $x = a$, so that what remains runs from $x = -a$ to $x = -a + t$. Work out V and A. The substitution $s = x + a$ helps.

Pause and try The ruling must hold after any number of slices have gone, so for every t with $0 < t \leq 2a$. What is the largest value of $\frac{V}{A}$ on this range?

STAGE 4 Beyond the question: is $2\frac{2}{3}$ enough?

Covers: beyond the question

Ask yourself Not asked: is your condition on a also enough? What if slices are eaten from both ends?

What's really going on

Archimedes' hat-box theorem

Archimedes proved, in *On the Sphere and Cylinder*, that the area of the band of a sphere between two parallel planes equals the area of the band they cut from the cylinder that wraps around the sphere (Figure 4.8.1). Our calculation is this theorem: the cylinder of radius a has crust area $2\pi a$ per unit length, and so does the sphere. Taking the whole sphere, its surface area equals that of the cylinder's curved surface, $2\pi a \times 2a = 4\pi a^2$. Archimedes regarded this, together with the fact that the sphere has two-thirds the volume of the cylinder, as his finest work. Cicero, visiting Syracuse in 75 BC, found the overgrown tomb by looking for the sphere and cylinder carved on it.

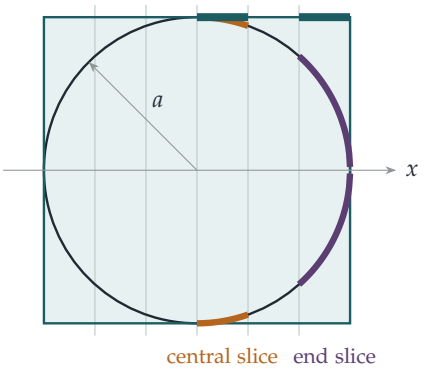


Figure 4.8.1 Cross-section of a spherical loaf of radius a inside its wrapping cylinder, cut into slices of equal thickness. The central slice (amber) has a short, nearly horizontal arc of crust far from the axis; the end slice (plum) has a long, steep arc close to the axis. When rotated about the x-axis, both give exactly the same crust area as the corresponding band of the cylinder (teal): $2\pi a$ times the thickness.

Maps of the Earth

The hat-box theorem gives a way to draw a map of the Earth that preserves area. Project each point of the globe horizontally outwards onto the wrapping cylinder, then unroll the cylinder. Bands of the sphere map to bands of the cylinder of the same area, and the same is true for any region. This is *Lambert's cylindrical equal-area projection* (1772). It distorts shapes badly near the poles, but areas are honest, unlike the familiar Mercator projection, which makes Greenland look as big as Africa (it is about fourteen times smaller). A famous theorem of Gauss, the *Theorema Egregium* (1827), shows that no map of a sphere can preserve both areas and shapes: every flat map of the Earth must distort something.

Random points on a sphere

Here is a practical consequence. To choose a point uniformly at random on a sphere, you might try choosing a latitude and longitude uniformly, but that crowds points near the poles. The hat-box theorem gives the right recipe: choose the height x uniformly between $-a$ and a, and the longitude uniformly between 0 and 2π. Because area is proportional to height, this is exactly uniform. Computer simulations in physics and graphics use this method. (Remarkably, it works only in three dimensions: the corresponding statement for spheres in four or more dimensions is false.)

Try this next

1 **The whole loaf.** Show that for the whole uncut loaf, $V/A = \frac{a}{3}$. Why is the condition in the question harder to satisfy than $\frac{a}{3} < 1$?

2 **The tropics.** What fraction of the Earth's surface lies between latitudes 23.4° south and 23.4° north? (Treat the Earth as a sphere. Since area is proportional to height, you only need $\sin 23.4^\circ$.)

3 **Other shapes.** Show that for a loaf in the shape of a paraboloid (rotate $y = \sqrt{x}$ about the x-axis), the crust per slice is *not* constant. For which curves $y(x)$ is $y\sqrt{1 + (y')^2}$ constant? (Solve the differential equation.)

4 **Volumes.** Use the volume formula to show that a sphere has two-thirds the volume of its wrapping cylinder, as Archimedes found.

Further reading

- **Sherman Stein, *Archimedes: What Did He Do Besides Cry Eureka?*** (Mathematical Association of America). Archimedes' mathematics explained for a modern reader.
- **Timothy Gowers, *Mathematics: A Very Short Introduction*** (Oxford University Press). The chapter on geometry introduces the geometry of curved surfaces.
- **John Snyder, *Flattening the Earth*** (University of Chicago Press). Two thousand years of map projections.

The full guided solution is in Section 4.8:
https://beyondstep.keithmcnulty.org/ch04/spherical-loaf.html